Now for (ii) let us calculate the limit of f as $(x,y)\to(0,0)$ along the line $y=x$. Then $\lim\limits_{(x,y)\to(0,0)} f(x,y)=0$; because along the line $y=x$ when $x\neq 0, f(x,y)=0$, But $f(0,0)=1\neq 0$; hence f cannot be continuous at $(0,0)$.

Example 8.12

Let $F(x,y)=x^3y+y^2x+7$ for all $(x,y)\in\mathbb{R}^2$. Calculate $\frac{\partial F}{\partial x}(-1,3)$ and $\frac{\partial F}{\partial y}(-2,1)$.

Solution

First we shall calculate $\frac{\partial F}{\partial x}(x,y)$, then we evaluate it at $(-1,3)$. As we have already observed, we find the derivative with respect to x holding y as a constant. That is,

$$\begin{aligned}\frac{\partial F}{\partial x}(x,y)=\frac{\partial(x^3y+y^2x+7)}{\partial x} &= \frac{\partial(x^3y)}{\partial x}+\frac{\partial(y^2x)}{\partial x}+\frac{\partial(7)}{\partial x}\\ &= 3x^2y+y^2+0\\ &= 3x^2y+y^2.\end{aligned}$$

So $\frac{\partial F}{\partial x}(-1,3)=3(-1)^2 3+3^2=18$.

Next similarly we find partial derivative with respect to y.

$$\begin{aligned}\frac{\partial F}{\partial y}(x,y)=\frac{\partial(x^3y+y^2x+7)}{\partial y} &= \frac{\partial(x^3y)}{\partial y}+\frac{\partial(y^2x)}{\partial y}+\frac{\partial(7)}{\partial y}\\ &= x^3+2yx+0\\ &= x^3+2yx.\end{aligned}$$

Hence we have $\frac{\partial F}{\partial y}(-2,1)=(-2)^3+2(1)(-2)=-12$.

Note that in the above example $\frac{\partial F}{\partial x}(x,y)=3x^2y+y^2$, which is again a function of two variables. So we can take the partial derivative of this function with respect to x or y. For instance, if we take $G(x,y)=3x^2y+y^2$, then we find $\frac{\partial G}{\partial x}=6xy$. Since $G(x,y)=\frac{\partial F}{\partial x}$, we have $\frac{\partial G}{\partial x}=\frac{\partial}{\partial x}\left(\frac{\partial F}{\partial x}\right)=6xy$.

We denote this as $\frac{\partial^2 F}{\partial x^2}$; which is called the second order partial derivative of F with respect to x.

Also, $\frac{\partial G}{\partial y}=3x^2+2y$. Since $G(x,y)=\frac{\partial F}{\partial x}$, we have $\frac{\partial G}{\partial y}=\frac{\partial}{\partial y}\left(\frac{\partial F}{\partial x}\right)=3x^2+2y$. We denote this as $\frac{\partial^2 F}{\partial y\partial x}$; which is called the mixed partial derivative of F with respect to x,y. Similarly we can also calculate $\frac{\partial}{\partial x}\left(\frac{\partial F}{\partial y}\right)=3x^2+2y$.

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